

Hands-on Workshop: Multiple particle tracking microrheology in planktonic systems

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Learning objectives

In this workshop you will:

- Understand the general principles of multiple particle tracking microrheology (MPTM)
- Explore conceptual differences between MPTM and bulk rheology
- Use a field viscometer (μ VISC) to measure bulk viscosity
- Prepare samples for microrheology experiments
- Acquire videos of microspheres undergoing Brownian motion
- Track beads and analyse trajectories to estimate rheological properties

General Workflow

The workshop is organized in three activities (see below) of increased complexity. Let's see how far we can go!

In all activities we will use the same basic steps of data collection and analyses. Here is a brief description of the main procedures.

μ rheology

To conduct microrheology we will use a technique called multiple particle tracking microrheology (MPTM), employing a hand-made fluorescence microscope and a camera to track beads suspended in our sample. For details on how to conduct MPTM refer to the accompanying document ('ViscosityMaps_protocol.pdf'). In general, the main steps to obtain and analyze videos for microrheology are:

- Seed 100 μ L of sample with 4 μ L of the beads suspension.
- Mount a slide carefully using an adhesive spacer
- Acquire videos (~1 min, 20 fps, 1024X1024 pixels) using the camera software (μ eye)

Data analysis

Data analysis will be performed with MATLAB, using the accompanying live script ('MOB_MPTM.m')

μVISC operation

- Load your sample into a syringe
(If sample contains >10 μm particles, filter)
- Run measurement X3 and record viscosity and temperature.
- Clean between samples by running mQ water.
- Remember to recalculate viscosity values using the calibration provided.

ACTIVITIES

Activity 1: Bulk vs Microrheology

Objective: We will compare viscosity measurements conducted with microrheology and bulk rheology on Newtonian fluids of different Ficoll concentration.

Steps:

1. Measure freshwater and at least 2 Ficoll solutions at different concentrations using the μVisc field viscometer.
2. Perform microrheology on the same solutions
 1. Prepare slides with Ficoll solutions
 2. Take 1 minute videos of the beads moving, at 20 fps frame rate.
 3. Analyse them with MATLAB to obtain mean viscosities
3. Create a scatterplot of the two kinds of measurements and fit a model

Discussion points:

- How do the two measurement approaches compare?
- What are the main sources of error?
- Can you estimate the temperature at the slide during measurements?

Activity 2: Microrheology of a diatom Culture

Objective: We will map viscosity around a diatom cell or within an aggregate.

Steps:

1. Prepare slides with a sample from the diatom culture, (as in the activity 1)
2. Take 1 minute videos of a population of suspended beads in or around an object of interest
3. Analyze them with MATLAB to obtain colormaps of relative viscosity

Discussion points:

- Are there any blank areas in the map? Why?
- How could we improve spatial coverage?

Activity 3: Viscoelasticity

Objective: we will apply these techniques to a viscoelastic solution to see how they can be used also to characterize complex behaviours.

Steps:

1. Prepare a slide as before with polyethylene oxide (PEO).
2. Analyse them with MATLAB to obtain mean viscosities
3. Compare MSD vs lag curves obtained with Ficoll solutions
4. Estimate viscoelastic properties

Discussion Points: Is this setup adequate for unveiling elastic behaviours?